

Seth Baird

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EDUCATION

- **Purdue University** West Lafayette, Indiana
Anticipated Graduation May 2027
Bachelor's in Computer Engineering Technology *Minor in Information Technology*

PROJECTS

- **Guitar Amplifier with Custom PCB** 12/2025
 - Engineered and fabricated a miniature guitar amplifier with a CAD-drawn case and a custom PCB in six weeks.
 - Simulated the circuit in Multisim and compared the frequency response curve against the one created in lab.
 - Directed a small team and utilized system diagrams, functional decomposition, Gantt, and critical path analysis.
- **Modernized Superheterodyne FM Radio Receiver** 04/2025
 - Redesigned an analog superheterodyne FM radio with digital automated channel scan, presets, and a display.
 - The implementation used DSP to run frequency calculations and communicated with the tuning DAC over I2C.
 - Extrapolated the characteristics of the VCO using quartic regression over 56 data points and an R^2 of 0.9987.
- **Engine Monitoring Android App** 02/2024
 - Developed a driver companion app to aid in learning how to use a manual transmission.
 - Aggregated engine parameters (OBD-II PIDs) using MQTT and processed this data within the app.
- **Replica Arcade Controller** 11/2022
 - Designed the HID hardware in OpenSCAD, allowing 4 variations of a part to be defined in a single file.
 - Deployed a Raspberry Pi Pico in gadget mode running rust firmware to handle user inputs and transmit it back.

SKILLS

C | Java | Python | Embedded Systems | RTOS | Linux | CAD | PCB Design | PLC | ISO 26262 | IEC 61131-3

EXPERIENCE

- **Purdue Electric Racing: Software Team**
 - Enhanced the vehicle's CANbus architecture by using two buses simultaneously for new high-speed motors.
 - Developed the PWM HAL (hardware abstraction layer) to control onboard variable-speed devices like cooling fans. This made it possible to modulate the speed in two lines instead of 50, promoting easier addition of new hardware.
 - Leveraged specialized signal analysis equipment such as CANable and logic analyzers to verify the stability of safety-critical systems.
- **SMS InfoComm (Wistron Corporation)** Dallas, TX
NVIDIA Server Repair Technician 05/2024 - 08/2024
 - **System Repair and Troubleshooting** Diagnosed, refurbished, and built NVIDIA HGX 8-GPU units.
 - **Hardware Validation and Analysis** Utilized proprietary diagnostic tools to identify errors and discrepancies within motherboard subsystems such as power delivery, signal integrity, and FPGAs before executing a repair.
 - **Team Workflow Optimization** Developed a Python script to automate thousands of lines of by-hand data entry into the company's antiquated database, saving my team 80 hours of labor per week.
- **Jawa.gg** Remote
Verified Seller 10/2021 - 08/2023
 - **Top 10 Seller in Overall Performance** Recognized by CEO Amanda Stefan for driving Jawa's growth.
 - **Computer Sales and Commissions** Built and sold dozens of tailored desktop, laptop, and Mac computers to end users at different price points and configurations.
 - **Part Refurbishment and Sales** Repaired, tested, and refurbished parts and sold them to PC builders and end users through Jawa's marketplace.

RELEVANT COURSEWORK

Advanced Embedded Systems | Prototyping & Development | Industrial Controls | Data Structures & Algorithms